

RAHUL MITRA

+65 8515 5413 | mitrasahul2002@gmail.com | linkedin.com/in/rahulmitra-dev | rahul-mitra.com | github.com/Rah-Rah-Mitra

EDUCATION

National University of Singapore

Aug 2023 – Jul 2027

Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Simulation, Stochastic Processes, Statistics for Engineering Applications, Operations Research, Artificial Intelligence, Computer Graphics, 3D Computer Vision, Software Engineering, Data Structures and Algorithms.

WORK EXPERIENCE

Operations Research (NUS System Design Project), Contract, STMicroelectronics

Aug 2026 – Present

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

Robotics Vision Engineer, Amazon (Blended AI+X)

Jul 2026 – Present

- Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.
- Improve processing and evaluation stages to upscale frames 4 times, lifting PSNR and SSIM so a low-cost camera reaches frame quality on par with premium sensors.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 – Present

- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.
- Convert 4 plants' Advanced Process Control models from Excel master files into a single HiGHS linear program returning setpoints, binding constraints, and shadow prices at millisecond latency.
- Built 20+ regression tests, a headless notebook runner, a deterministic model stand-in for GPU-free CI, and committed golden snapshots for reproducible runs.

Platform & Solutions Engineer (Sparks Citizen Developer), People's Association

Sep 2025 – Sep 2026

- Delivered a digital garden-plot balloting platform across admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Led deployment planning, multi-repo coordination, and stakeholder management across product, engineering, and operations, winning the S\$20,000 Sparks Community Innovation Fund.

Data Analytics & Product Management Intern, Abbott Laboratories

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Productionized a team-built APC simulator for live internal users on Azure App Service with Docker packaging, access restrictions, and runtime and session hardening.

PROJECTS AND COMPETITIONS

Hackathon Build, OnTheSpectrum: AI Engineer Hackathon 2026

May 2026

- Created a game of AI-driven infinite levels in a 7-hour hackathon build, wiring Codex and Blender MCP into a local-first Blender-to-Three.js pipeline for GLB assets, previews, and playable-world QA.

Personal Project, Hailo Edge AI Training and Deployment

Mar 2026

- Trained YOLO26 through transfer learning and reshaped its architecture with the Hailo Dataflow Compiler, running document detection on a Hailo-8L rated 13 TOPS at 1.5 W to speed up large-batch scanning.

Hackathon Project, Waaah Comics: 24-Hour Gesture-to-Comic Build

Jan 2026

- Turned MediaPipe 3D hand landmarks into generated comic strips in a 24-hour sprint with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.

- Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)** **Jan 2026**
- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.
- Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon** **Jun 2025**
- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.
- AI/ML Engineer, AgeWellLah.AI: NUS HealthHack** **May 2025**
- Developed an elderly-care triage chatbot on OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.
- Hackathon Project, SmartExam: GovTech x NTUPC Product Hackathon** **Apr 2025**
- Streamlined exam practice into a multi-agent RAG pipeline turning students' own notes into adaptive-difficulty papers, replacing hundreds of dollars of past-year paper purchases.
- Competitor, DSTA BRINHACK 2025 CTF** **Feb 2025**
- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, finishing around 90th of 400+ teams.

LEADERSHIP AND ACTIVITIES

- YesWeHack, Independent Bug Bounty Researcher** **May 2024 – Present**
- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.
 - Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports with Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.
 - Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.
- NTUC Health Active Ageing Centre, Elderly Care Volunteer** **Sep 2025 – Sep 2026**
- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.
- Republic of Singapore Navy, Base Support Assistant** **Jan 2022 – Jan 2023**
- Compressed a 6-month food-waste analysis cycle into a 3-minute reporting process with an Excel VBA automation workflow inside a security-constrained environment.
 - Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.
- STEM Inc. (Anderson Serangoon Junior College), Astronomy Head & Robotics Member** **Mar 2019 – Aug 2020**
- Led astronomy outreach, competed in Singapore's 7th Astronomical Olympiad, and engineered an autonomous robot for the National Robotics Challenge.

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA

Web and Cloud: React, Next.js, Three.js, FastAPI, Flask, Docker, AWS (Fargate, Route 53), Terraform, Azure App Service, Redis, Kafka, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping

AI and Machine Learning: RAG, AI agents, vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), CUDA/HPC on ASPIRE 2A

AI Foundations: informed search and heuristics, constraint satisfaction, adversarial search, logical inference and resolution, probabilistic reasoning, Bayesian networks

3D Vision: camera models and calibration, homography estimation, single-view metrology, epipolar geometry (fundamental and essential matrices), absolute pose estimation (PnP), three-view geometry, generalised camera models, structure-from-motion with bundle adjustment, two-view and multi-view stereo

Simulation and Operations Research: discrete-event simulation, input modelling, output analysis, verification and validation, random variate generation, queuing systems, SimPy, OR-Tools CP-SAT, HiGHS linear and quadratic programming, hybrid flow-shop scheduling

Statistics: hypothesis testing, confidence and prediction intervals, maximum likelihood estimation, linear regression, ANOVA, bootstrap methods, goodness-of-fit tests

Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt); Reverse Engineering Windows Executables; Excel 2019 Advanced (Intellecty)